

# Network Optimisation

## Contents

|                                              |                                     |
|----------------------------------------------|-------------------------------------|
| Goal of the project .....                    | 2                                   |
| Optimization.....                            | 2                                   |
| Data for the algorithm.....                  | 4                                   |
| Input.....                                   | 4                                   |
| Structure .....                              | 4                                   |
| Description of Proforma.....                 | 5                                   |
| Output.....                                  | 6                                   |
| Calculation rules .....                      | 7                                   |
| Not perfect calculations .....               | 8                                   |
| What is not included from the Proforma ..... | 8                                   |
| Allocation .....                             | <b>Error! Bookmark not defined.</b> |
| Cost .....                                   | 8                                   |
| Bunkering Costs (Weekly) .....               | 9                                   |
| Chartering Costs (Weekly) .....              | 8                                   |
| Port Call Costs.....                         | 15                                  |
| Speed .....                                  | <b>Error! Bookmark not defined.</b> |
| Port Productivities.....                     | 15                                  |
| Wait Times (+ Delay Valorisation).....       | <b>Error! Bookmark not defined.</b> |
| Man Times (Man in/out) .....                 | <b>Error! Bookmark not defined.</b> |
| Transshipment Dwell Time Constraint .....    | <b>Error! Bookmark not defined.</b> |
| Transit Time.....                            | <b>Error! Bookmark not defined.</b> |
| Line Buffer (wait + sea slack) .....         | <b>Error! Bookmark not defined.</b> |
| Frozen Lines.....                            | <b>Error! Bookmark not defined.</b> |
| Data ingestion .....                         | <b>Error! Bookmark not defined.</b> |
| Input files.....                             | <b>Error! Bookmark not defined.</b> |
| Transformation .....                         | <b>Error! Bookmark not defined.</b> |

**Remark:** The project is done in an iterative way where more constraints can be added to make the results easier to apply in practice.

## Goal of the project

Determine the best network

- to service the current demand
- for the lowest cost
- while maintaining acceptable POL-to-POD cargo transit times
- starting from the existing network
  - the goal is not to start from scratch but found the top changes in the current network that would lower the cost the most

Lines are designed to be weekly. The network being made of lines is also designed to be weekly. The data being used is monthly

- Proforma from previous month
- Cargo flow from previous month

## Optimization

- Start from existing network
- Objective
  - Minimise cost (calculation explained in the sections below)
    - Chartering
    - Bunkering
    - Port call cost
    - Transhipments
- Constraints
  - Base constraints
    - Satisfy all demand (soft)
    - Vessel capacities
    - Respect cargo transit time (soft)
      - Cargo value fully depreciates if the route it takes is tardy (by a fraction of its expected transit time)

- Max. number of operations performed at each port is upper-bounded by (port stay time/productivity)
  - Vessel speed must be between 10kts (hard) and 16.5kts (soft)
    - Steaming above 16.5kts possible, but heavily penalised
- Additional constraints for lines
  - Line will be penalised if the buffer is above 30%
  - Number of ports in a Service that are called more than once  $\leq 2$
  - Minimum 3h berthing time
  - A route cannot come in and out of APAC
  - Number of times a port can be visited in the same Service  $\leq 3$
  - Max direct port to port connection = 2k miles
  - Every line must visit at least one TS port
  - Number of unique ports in a service  $\leq 20$ 
    - We can force longer or shorter services to explore new ideas
  - All POL-POD pairs (in the demand) must be connected by no more than 3 transshipments
  - No repeating of sub-routes within a service's port rotation
    - E.g. CNXMN – TWKHH – TWTXG – CNXMN – TWKHH not allowed
  - Frozen lines
    - These lines are not fully operated by CNC and cannot be modified.
      - Port rotation and speed can't be modified
      - Flow on them can be modified.
- Additional constraints for flow
  - Cargo can be transshipped only at select ports (including ports that carried out intra-Asia transshipments in the demand routing data)
  - There needs to be at least 1 day difference between ETB2 and ETD1 for a transshipment between 2 lines to be possible
  - Singapore transshipments must be at least 72.5% of SG/MY/ID transshipments
- Actions Possible
  - Addition of a regular/transshipment port call on a service
    - Option 1:  $A \rightarrow B \rightarrow C$  becomes  $A \rightarrow B \rightarrow X \rightarrow C$
    - Option 2:  $A \rightarrow B \rightarrow C$  becomes  $A \rightarrow B \rightarrow X \rightarrow B \rightarrow C$
  - Removal/Change of a regular/transshipment port on a service

- Shifting a port call from one service to another (can also be done on a single service, i.e. “sliding” the position of the port call)
- Swapping the exact positions of 2 port calls (can be done between distinct lines, or on the same line)
- Change vessel size rank
- Change vessel speed between port
- Change the round-trip duration (and number of vessels deployed) of the service

## Data for the algorithm

### Input

### Structure

| DEMAND |                             |
|--------|-----------------------------|
| TABLE  | COLUMNS                     |
| Demand | POL (Key)                   |
|        | POD(Key)                    |
|        | Expected Transit Time (Key) |
|        | Weekly Volume               |

| PORTS (ASIA) |                        |
|--------------|------------------------|
| TABLE        | COLUMNS                |
| Port         | Port ID (Key)          |
|              | Longitude              |
|              | Latitude               |
|              | Transshipment Cost     |
|              | Transshipment Capacity |

| PORT PARAMETERS |                        |
|-----------------|------------------------|
| TABLE           | COLUMNS                |
| Port parameter  | Port ID (Key)          |
|                 | Vessel Size Rank (Key) |
|                 | Cost                   |
|                 | Productivity           |
|                 | Waiting Time           |

| PORT MAN TIMES |               |
|----------------|---------------|
| TABLE          | COLUMNS       |
| Man Time       | Port ID (Key) |
|                | Man IN        |
|                | Man OUT       |

| VESSEL PARAMETERS |                              |
|-------------------|------------------------------|
| TABLE             | COLUMNS                      |
| Vessel            | Vessel Size Rank (Key)       |
|                   | Vessel Size Class            |
|                   | Nominal Capacity             |
|                   | Daily Charter Cost           |
|                   | Bunker Consumption (Man)     |
|                   | Bunker Consumption (Port)    |
|                   | Bunker Consumption (Sea Aux) |

| SAILING BUNKER CONSUMPTION |                              |
|----------------------------|------------------------------|
| TABLE                      | COLUMNS                      |
| Vessel                     | Vessel Size Rank (Key)       |
|                            | Speed (Key)                  |
|                            | Bunker Consumption (Sailing) |

| DISTANCES (CNC) |                  |
|-----------------|------------------|
| TABLE           | COLUMNS          |
| Distance        | POL (Key)        |
|                 | POD(Key)         |
|                 | Sailing Distance |

| LINES (GENERAL) |                     |
|-----------------|---------------------|
| TABLE           | COLUMNS             |
| Lines           | Line Name (Key)     |
|                 | Vessel Size Rank    |
|                 | Nominal Capacity    |
|                 | Round-trip Duration |
|                 | Service Type        |
|                 | Ignore Buffer LB    |

| PROFORMA |                         |
|----------|-------------------------|
| TABLE    | COLUMNS                 |
| Proforma | Line Name (Key)         |
|          | Rotation Sequence (Key) |
|          | Port ID                 |
|          | EOSP                    |
|          | Wait Time               |
|          | Man IN                  |
|          | Ops Time                |
|          | Man OUT                 |
|          | Time to Next            |
|          | Speed                   |
|          | Capacity Scale          |
|          | Effective Capacity      |
|          | Capacity Reserved       |
|          | Allocation              |

### Description of Ports

- Transshipment capacity is an indicator of whether the port is a transshipment port
- Productivity represents the amount of cargo movements that can be performed in an hour.

## Description of Lines and Proforma

Example of a line's proforma (shows planned schedules and operations):

| linename | portid | sequence | eosp_utc_wd | eosp_utc_hr | time_wait | time_manin | staytime | time_manout | timetonext | vspeed | cap_nom | vrank | cap_eff | duration | moves | ops_prod | alloc | cap_scale | cap_reserve | svc_type | ignore_buffer_lb |
|----------|--------|----------|-------------|-------------|-----------|------------|----------|-------------|------------|--------|---------|-------|---------|----------|-------|----------|-------|-----------|-------------|----------|------------------|
| BBX2CNC  | CNGB   | 1        | 1           | 17          | 26        | 3          | 14       | 3           | 11         | 14     | 2822    | 5     | 2547    | 35       | 800   | 57       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | CNSHA  | 2        |             |             | 25        | 4          | 12       | 4           | 46         | 13.8   | 2822    | 5     | 2547    | 35       | 1100  | 92       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | TWKHH  | 3        |             |             | 1         | 2          | 13       | 1           | 129        | 14     | 2822    | 5     | 2547    | 35       | 500   | 38       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | MYPKG  | 4        |             |             | 6         | 1          | 12       | 1           | 82         | 16     | 2822    | 5     | 2547    | 35       | 550   | 46       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | BDCGP  | 5        |             |             | 36        | 2          | 70       | 2           | 100        | 13.1   | 2822    | 5     | 2547    | 35       | 1550  | 22       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | MYPKG  | 6        |             |             | 17        | 1          | 30       | 1           | 55         | 14.4   | 2822    | 5     | 2547    | 35       | 1500  | 50       | 1812  | 0.90255   | 735         | OWN      | 0                |
| BBX2CNC  | VNSGN  | 7        |             |             | 7         | 4          | 12       | 3           | 104        | 14.3   | 2822    | 5     | 2547    | 35       | 680   | 57       | 1812  | 0.90255   | 735         | OWN      | 0                |

Every line goes through the same exact states (in order) at each port:

- EOSP represents the time of week (in UTC) the line reaches the port
  - o **eosp\_utc\_wd** is a number (0-6) representing the EOSP's day of week (0 is Monday, 1 is Tuesday, 6 is Sunday)
  - o **eosp\_utc\_hr** is a number representing the hour of day of the EOSP
  - o Sufficient to provide the first port's EOSP, since the day and time of week (at any port and stage) is computable from any given EOSP
- The line waits at the port before berthing
  - o **time\_wait** represents (in hours) an average waiting time for the line (or vessel of similar size) before berthing. Sometimes, this wait time includes an artificial buffer in case more time at port is required for cargo movements
- **time\_manin**
- The line is fully berthed, and operations (loading/unloading) can start
  - o **staytime** represents (in hours) the amount of time the vessel is allocated to perform the cargo movements.
  - o **ops\_prod** represents the planned productivity of the line at the port (in # movements per hour), according to the proforma
    - For actual productivity used in calculations, refer to the section "Port Productivities"
- **time\_manout**
- The line departs and sails to the next port
  - o **timetonext** represents (in hours) the sailing time to the next port
  - o **vspeed** represents (in knots) the average speed of sailing between the current and next port

Other fields that relate to:

- Capacity of a line:
  - o **cap\_nom** represents (in TEUs) the capacity that the vessel is rated for, regardless of cargo weight
    - If we want to modify the size of the vessel, we refer to **cap\_nom** in the "Vessel\_Nominal" dataset for the corresponding new vessel rank
  - o **cap\_eff** represents (in TEUs) the maximum capacity allowed on the vessel (after accounting for the service's expected trade weight)
    - After applying **cap\_scale** to **cap\_nom**
  - o **alloc** represents (in TEUs) the actual available capacity for CMA/CNC cargo to be transported on
    - After deducting **cap\_reserve** from **cap\_eff** (after removing slots that are swapped/sold to partner services)

- Service type (and modifiability of a line):
  - o **svc\_type** takes on 3 values:
    - **VSA** – service is not fully-operated by CNC; we are allowed to reroute cargo meant for these lines, otherwise immutable (no optimisation wrt speed/port rotation/vessel size)
    - **FIX** – service fully-operated by CNC but is highly specialised; port rotation is fixed, but optimisation wrt speed/vessel size and cargo (re)routing are allowed
    - **OWN** – all other lines. All sorts of optimisation are allowed on the line.
- Ignore Buffer LB: see buffer calculation for more details.

## Output

The results/proposed changes to the current network are to be provided in the following 3 files.

## Lines

Output proforma (see [\*output\\_proforma.xlsx\*](#) for reference) should be an excel file – each worksheet represents each sequential modification made to the existing network, and will show:

- Line name, port rotation sequence, port ID (key)
- EOSP time stamps (date-time of week the line arrives at the port before waiting)
- Timings (wait time, man IN, ops time, man OUT, sail time) in hours
- Allocation (capacity of line) and amount of cargo flowed on the particular leg
- How many movements were made at port (moves):
  - o **mvt\_load** – amount of cargo loaded on the line at the port (POL)
  - o **mvt\_disch** – amount of cargo discharged from the line at the port (POD)
  - o **mvt\_transload** – amount of transhipped (and dwelling) cargo reloaded on the line at the port (PTS)
  - o **mvt\_transdisch** – amount of transhipped (to start dwelling) cargo discharged from the line at the port (PTS)

## Cargo flow

Cargo flow (see [\*output\\_routings.xlsx\*](#) for reference) should be an excel file.

The first worksheet shows the actual routing of cargo, indexed by:

- POL and POD of the demand
- Expected transit time of the demand
- Cargo path that is taken (forfeited cargo will take on a dummy “\_\_FORFEITED\_\_” path). For every sequential modification made to the existing network, the cargo flow table should show:
  - o Routed demand:
    - Volume of the demand routed along that path
    - Actual transit time of the cargo path (from its corresponding proforma)

- Amount of delay (if applicable)
- Forfeited demand:
  - Volume of demand forfeited

The second worksheet should show, at every sequential modification made to the existing network, how many total transshipments are made.

- We will also want to track the number of intra-Asia transshipments made at certain TS ports of interest (CNSHA, CNSHK, HKHKG, KRPUS, MYPKG, SGSIN)

## KPI

KPI output (see [\*output\\_summary.xlsx\*](#) for reference) should be an excel file – each worksheet represents each sequential modification made to the existing network, and summarises each line’s rotations, times, and the following fields:

- FF (filling factor):
  - Represents how “full” the line is (in %) for each individual sailing leg (essentially cargo flow divided by allocation)
  - maxFF is just the maximum FF achieved by the line over the entire rotation
- Weekly Costs:
  - Bunkering Costs: the sum of bunkering costs (consumption \* **579** USD/tn) for
    - Sea Aux, Berth, Man (linear in duration; see below) and
    - Sea Main (sailing) – use the interpolation method.
  - Chartering Costs: see below
  - Port Call Costs: see below
  - Transshipment Costs: 50/50 split amongst the participating lines
    - Note that this isn’t a cost that is internalised by the liner services
- Buffer: see below

There should also be a grand sum (across all lines) for:

- Number of vessels deployed
- Number of port calls made in a week
- Total system cost (and its constituent costs summed across every line)

## Calculation rules

The calculation is done using the same formula as the line simulation tool based on the proforma with some differences.

The goal of the calculation is mostly to determine the system cost (total cost) of the network with

- Port call cost
  - The cost paid when calling a port

- Chartering cost
  - The cost paid when renting a vessel
- Bunker cost
  - The cost to operate the vessel
  - This is mostly linked to the speed of the vessel, so the speed calculation will fall under this cost.

## Not perfect calculations

Our calculations can't be as perfect as reality because there is no knowing what will be the exact vessel deployed on a line if a line adds or removes a vessel. Depending on the age of the vessel or the time when it is chartered, the cost of operating it can vary a lot. It was then decided to do the calculation at vessel class level by averaging the cost of all the vessels in the class. This is not such an issue as the main idea is to compare the original network and to see if the new suggested one is cheaper while performant.

## What is not included from the Proforma

These costs are not accounted for in our algorithm:

- Tonnage Dues
- Admin Call Costs

For vessels, only VLSFO engine will be used

## Cost

The cost is calculated at weekly level.

## System cost

$$System\ cost = Cost_{charter} + Cost_{portcall} + Cost_{bunker} + Cost_{transhipment}$$

## Chartering Costs

$$Cost_{charter} = p_{internal} * 7 * n_{vessels}$$

For a service, we take the average (per-day) charter cost  $p_{internal}$  across all vessels similar in size (vessel rank) of the vessel deployed on this line, and scale it accordingly.

## Bunkering Cost

The overall bunkering cost is given by.

$$Cost_{bunker} = \text{bunkering consumption} * \text{bunker price}$$

The bunkering mostly depends on the speed of the vessel and the time spent navigating compared to the time spent at port discharging containers. We will continue to use the unit bunker price of 579 USD/tn.

### *Bunkering consumption*

The bunkering consumption is the sum of 4 different consumption parts.

$$\text{bunkering consumption} = \text{Berth} + \text{Man} + \text{Sea Aux} + \text{Sea Main}$$

### Sea Main

The consumption of the main motor at sea, meaning between ports.

Bunker consumption rate of a vessel is nonlinear with sailing speed. As such, we've computed the individual bunker consumption rates for different vessel size ranks for speeds ranging from 10kts to 18.5kts inclusive (at 0.5kts discretisation).

Currently, we are mapping speeds (and their corresponding bunker consumption rates and costs) to sail durations and linearly interpolating between the pre-defined segments. For example:

- Sailing distance between SGSIN and CNSHA is 2142.84 NM. I am using a vessel of size rank 5.
  - o At 15.0kts: consumption rate is 32.71 tns/day; multiply by sail duration of 5.95 days for a total sailing consumption of 194.69 tns
  - o At 15.5kts: consumption rate is 35.38 tns/day; multiply by sail duration of 5.76 days for a total sailing consumption of 203.8 tns
- Current method of interpolation – if I were to want a sailing duration of 5.85 days, my bunker consumption would be obtained from a linear interpolation between the two support points (5.76 days, 203.8 tns) and (5.95 days, 194.69 tns).

Actual (nonlinear) formula:

The bunkering consumption is the consumption rate multiplied by the sea time.

$$Sea\ Main = sea\ time * \beta_{sea}(v)$$

The consumption rate of a particular vessel (with vessel parameters  $p_{(\cdot)}$ ) is modelled as a function of vessel speed:

$$\beta_{sea}(v) = (1 + p_{vessel}) e^{p_{general}} p_{s.draft}^{p_{draft}} \cdot v^{p_{speed}}$$

A service's consumption rate (between consecutive port visits) is determined by applying the formula to all vessels similar in size (vessel rank) and taking the average.

### Sea Aux

The consumption of the auxiliary motor at sea, meaning between ports.

$$Sea\ Aux = sea\ time * A/E\ at\ sea$$

A/E = Auxiliary bunkering at sea, where the consumption rate is constant (tons per day).

### Berth

The consumption while berthing in the port.

$$Berth = berth\ time * A/E\ in\ port$$

A/E in port = Auxiliary bunkering at port, where the consumption rate is constant (tons per day).

### Man

The consumption while manoeuvring in the port.

$$Man = Man\ time * (A/E\ Man + M/E\ canal)$$

A/E Man = Auxiliary bunkering while manoeuvring, where the consumption rate is constant (tons per day).

M/E Canal = Auxiliary bunkering also used for manoeuvring.

### Time

There are 3 times to be calculated

- Man time

- Berth time
- Sea time
  - That will give the speed based on distance

The starting point of the optimisation is the existing network.

The current network has already defined time at port, productivity that are dependant of the current situation at the port. This is done by human operators.

When a new port is added to a line, the values taken is from the historical data.

This is the logic to get the different time value

- for a new port being added
- for a different vessel class being deployed on an existing port in a line.

### Logic for modifying an existing service

This is the current idea we are using to populate the data fields (mostly timings) when we modify the services. Other than times taken from historical data (productivities, wait times, man times, etc) which are immutable, you may modify port stay time and sailing speed as you wish.

#### *Adding a port*

If we add a port (or swap an existing port call for another), the following fields are required:

- Wait time, man in/out time, port productivity
  - Take from historical data
- Stay Time (for moves)
  - Currently set to 6 hours (twice the minimum port stay time)
- Round-trip duration and speed
  - From existing round-trip duration, compute the total sailing time allowed; speed is averaged
  - We will add 7 days to the round-trip duration (and then recompute speed) if
    - Computed speed is too high (for now we use > 17.5 kts)
    - Computed speed is “negative” (i.e. no time allowed for sailing)
  - If average speed is < 10kts, we set the sailing speed to 10kts

- This means that there is more time allowed for port operations
- Scale up all port stay times to make up for this difference

### *Removing a port*

If we remove an existing port, we need only compute:

- Round-trip duration and speed
  - From existing round-trip duration, compute the total sailing time allowed; speed is averaged
  - If speed < 10kts (and sailtime >= 7 days)
    - We attempt to subtract 7 days from the round-trip duration (to speed up the line). Recompute speed
  - If speed is > 17.5kts
    - We attempt to add 7 days to the round-trip duration (to slow down the line). Recompute speed
  - If average speed is < 10kts, we set the sailing speed to 10kts
    - This means that there is more time allowed for port operations
    - Scale up all port stay times to make up for this difference

### *Changing vessel rank*

Modifying the vessel rank of a service affects, for all ports:

- Wait time and productivity
  - Take from historical data
- Capacity: see below

### Computing Allocation

To compute allocation (maximum # TEUs allowed to be routed on the line) from the nominal capacity of a service, we use the following formula:

$$Alloc = Cap_{nom} * cap\_scale - cap\_reserve$$

For existing (unmodified) lines, the nominal capacities (as well as allocations) are exactly as given in the proforma. If we wish to modify the vessel size of the line, we take its corresponding nominal capacity from the vessel dataset.

Parameters cap\_scale and cap\_reserve are line-specific, and are used in the following manner:

- Nominal capacity states how many TEUs the vessel is rated for;
- Applying cap\_scale to the nominal capacity gives us the effective capacity, which represents the amount of TEUs the line can actually carry (given the historical trade weight);
- Deducting cap\_reserve (amount of slots reserved for other purposes e.g. partner cargo) from effective capacity gives us the **allocation of the service – the maximum TEUs of CNC cargo allowed on the line**

## Speed

$$vessel\ speed = \frac{vessel\ traveling\ distance}{24 * sea\ time}$$

With  $vessel\ traveling\ distance = sum(sea\ distance\ between\ ports)$

## Line Buffer (wait + sea reserve)

For a line, the buffer is defined as the proportion of the following sum:

1. Total time spent waiting at port (**time\_wait** in proforma), and
2. The total sea reserve time, i.e. how much time can be saved if the vessel sails at its maximum allowed speed of 16.5 kts, instead of the prescribed speed,

out of the total port rotation duration (typically 7 days for each vessel deployed on the rotation).

The formula is given by:

$$Buffer = \frac{\sum_{i \rightarrow j} \left( t_i^{wait} + dist_{i \rightarrow j} * \left( \frac{1}{v_{i \rightarrow j}} - \frac{1}{16.5} \right) \right)}{24 * 7 * n_{vessels}}$$

We are currently constraining each line's buffers to be between 15% and 30%. For lines whose buffers (calculated from proforma) are already less than 15%, we drop the lower-bound buffer constraint, while keeping the upper-bound.

## Transshipment Dwell Time Constraint

For cargo routes involving transshipment, e.g. BDCGP/BBX3/**SGSIN**/BBX2/CNSHA, SGSIN is the port of transshipment (PTS). The transshipment dwell time is then defined as the amount of time the cargo is stored at the PTS (SGSIN) between the unloading line (BBX3) and the reloading line (BBX2), and is computed as follows:

$$t_{diff} = DTOW_{ops\_start}(BBX2, SGSIN) - DTOW_{ops\_end}(BBX3, SGSIN)$$

$$t_{dwell} = t_{ts}^{min} + \left( (t_{diff} - t_{ts}^{min}) \bmod 7 \right)$$

(\* DTOW stands for date-time of week, e.g. Mon 18:30)

The first line computes the dwell time conservatively – at the point in time where the unloading line (BBX3) has finished all operations, the cargo (to be transhipped) must be at the PTS already. Similarly, we must assume this cargo is ready to be reloaded to the next line as soon as the reloading line (BBX2) begins its operations at the PTS.

The second line ensures a minimum dwell time  $t_{ts}^{min}$  (assumed to be sufficient for cargo to be moved between terminals at the same port). For our calculations, we use a minimum dwell time of 1 day, meaning  $t_{dwell} \in [1, 8)$ .

## Transit Time

Port-to-port transit times for direct cargo routes, e.g. BDCGP/BBX3/SGSIN, is measured from the start-of-operations time at the POL till the end-of-operations time at the POD.

For cargo routes involving transshipment, e.g. BDCGP/BBX3/SGSIN/BBX2/CNSHA, we first split it up into direct cargo routes BDCGP/BBX3/SGSIN and SGSIN/BBX2/CNSHA. We then compute the total transit time as the sum of the individual port-to-port times and the transshipment dwell times.

For the abovementioned example, we have the transit time (POL start-of-ops till POD end-of-ops) calculated as the sum of:

- PTP1: Start-of-ops (BBX3, POL) to end-of-ops (BBX3, PTS)
- Dwell: End-of-ops (BBX3, PTS) to start-of-ops (BBX2, PTS)
- PTP2: Start-of-ops (BBX2, PTS) to end-of-ops (BBX2, POD)

The definition of direct port-to-port times and transshipment dwell times are defined such that the computation of transit times for transhipped cargo does not double-count or omit any portion of the cargo's route.

